



## The Icynene Insulation System®



**ICYNENE LDC-50**

**INFORMATION KIT**

**ERA POLYMERS PTY. LTD.** is the importer and distributor of The Icynene Insulation System<sup>®</sup>. Introduced into Australia in late 2007, Icynene has been prominent in North America for over 20 years. In that time it has risen to become the leading soft foam insulation in that market.



## ***Insulation like no other!***

The Icynene Insulation System is a soft, expanding foam thermal insulation and air barrier in one product. Because it is sprayed in place, it provides a custom fit every time, expanding around cavities and pipes, blocking leaking air, energy loss, dust, and noise.

Icynene sticks to any surface without the need for meshes, wires or plugs. It leaves no air pockets for moisture to condense. It is completely inert and does not affect any building materials. It can be used under roofs without sarking, walls without building wrap, and under floors without vapour barriers, stopping dampness filtering through to your feet.

Icynene works as well in real life installations as under laboratory testing. That means it will give your buildings decades of full quality insulation. Because it is water-based, there are no harmful chemicals and gases that can be associated with other insulations.

## ***Properties and Features***

- Complete insulation and air barrier in one product (R-2.6 / 100mm for LD-C-50)
- Up to 50% more energy savings
- Significant reduction in a home's Carbon Footprint
- No air leakage; fills all gaps and holes
- Effective noise control barrier
- Requires no sarking or building wrap\*
- Repels water, controls water vapour
- No mould or mildew growth
- Reduces Air Conditioning load and size
- Seals a house against dust and pollens
- No itchy, dusty, unhealthy fibres
- No 'cold floor' syndrome
- Sticks to any building surface
- Soft, light and flexes with your building
- No sagging or collapse in walls, floors or roofs
- Effective for the life of your building!

\* except in extremely humid, cold conditions, or where regulated



## An airtight solution for energy-efficient design and construction

With double-digit percentage increases in energy costs, there's a growing sense of urgency for the building community to find a better approach to energy efficient buildings.

Expanding knowledge of the building envelope offers valuable clues to creating more energy-efficient buildings and happier building owners and occupants. There's no better place to capitalise on this knowledge than in our approach to R-value and insulation performance.

The building industry has grown comfortable with the concept that more insulation is better, along with higher R-value, but product options like The Icynene Insulation System® offer a better approach to energy efficiency.

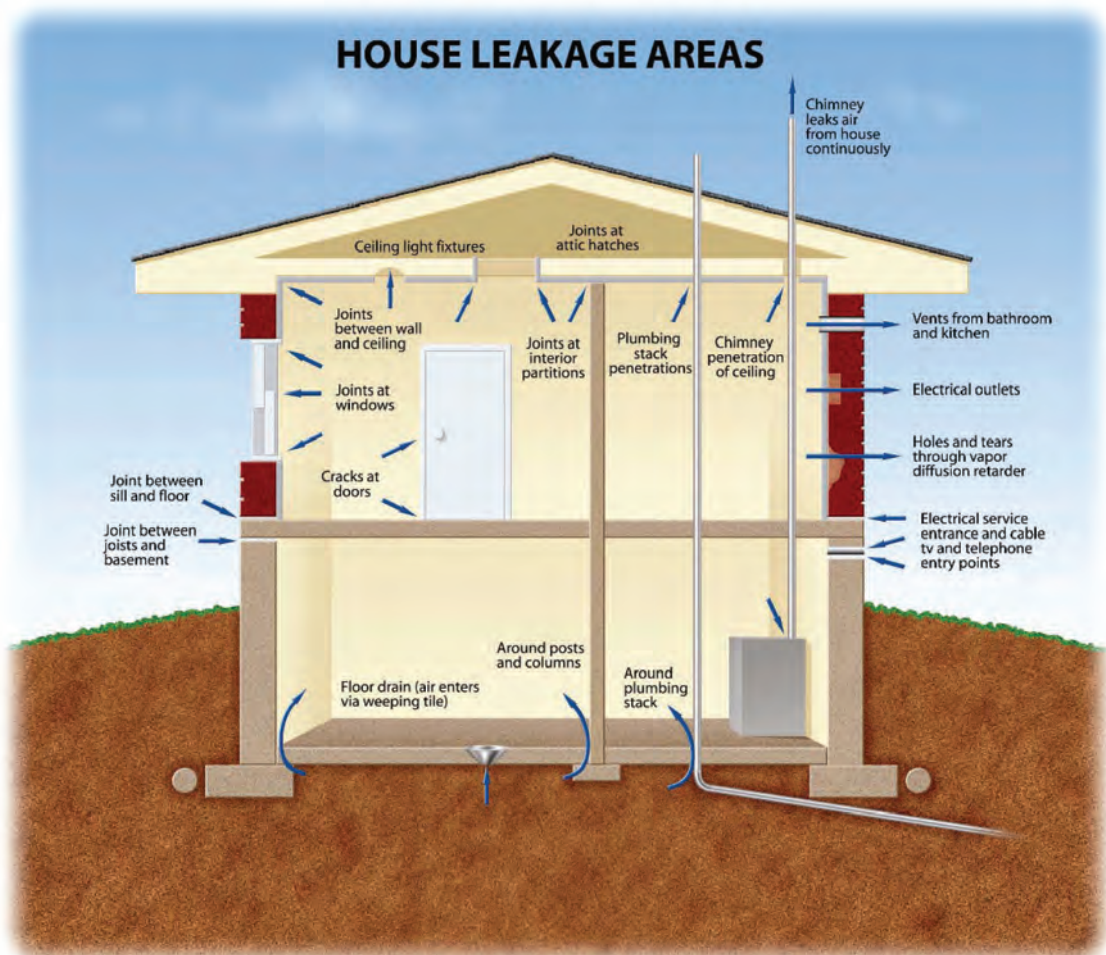
### Consider:

R-value can't do a thing about air leakage

Higher R-value provides minimal and diminishing returns

R-value measures an insulation's ability to inhibit conductive heat flow, but can't do a thing about the primary method of heat transfer - air leakage (convection) - which causes up to 50% of building energy loss (U.S. Department of Energy's Oak Ridge Laboratory).

93% of conductive heat flow is stopped by R-2.29 insulation. Upgrading from R-2.29 insulation to R-5.64 insulation, for example, reduces conductive heat flow by only another four percent (Fourier's Law of Thermodynamics).



## PRODUCT SPECIFICATION

### 1. PRODUCT NAME

# ICYNENE LD-C-50™

ICYNENE LD-C-50™ is a trademark for light density, open celled, flexible, 100% water-blown polyurethane foam insulation manufactured by Icynene Inc. ICYNENE LD-C-50™ is a nominal 8 kg/m<sup>3</sup> (0.5 lbs/ft<sup>3</sup>) density, free rise material.

### 2. MANUFACTURER

ICYNENE LD-C-50™ is made on-site from liquid components manufactured by Icynene Inc. Installation and on-site manufacturing is supplied by independent Icynene Licensed Dealers.

### 3. PRODUCT DESCRIPTION

ICYNENE LD-C-50™, the "classic" light density formulation of Icynene, has been installed in buildings since 1986. Icynene is the pioneer of high yield, 100% water-blown polyurethane foam technology for air-sealing and insulating buildings.

ICYNENE LD-C-50™ insulates and air-seals in one step for maximum energy conservation while minimizing the environmental impact during manufacturing and construction. Significantly reducing air leakage means ICYNENE LD-C-50™ contributes to a healthier, quieter and more comfortable indoor environment, while reducing energy consumption and related greenhouse gas emissions by as much as 50%.

ICYNENE LD-C-50™ is an effective vapour permeable air barrier material that can move with the building to maintain the air barrier characteristic against energy-robbing air leakage for the life of the building. Convective air movement inside wall cavities is virtually eliminated, providing more uniform temperatures throughout the building.

The result is superior quality construction, with higher comfort levels and lower heating and/or cooling costs. Energy savings will vary depending on building design, location, etc.

ICYNENE LD-C-50™ is applied by spraying liquid components onto an open wall, crawlspace, ceiling surface or cathedral ceiling. There it expands approximately 100:1 in seconds to provide a flexible foam blanket of millions of tiny air cells, filling building cavities, cracks and crevices in the process. It adheres to most construction materials, sealing out air infiltration. Excess material is easily trimmed off, leaving a surface ready for drywall or other code-compliant finish

### 4. TECHNICAL DATA

(Based on Core Samples)

#### Thermal Performance

Thermal resistance (ASTM C518):

- RSI/25.4 mm @ 24°C  
= RSI 0.65 m<sup>2</sup> °C/W
- R/in @ 75°F  
= R 3.7 hr. ft<sup>2</sup> °F/BTU

Average insulation contribution in a full fill stud wall:

- 38mm x 89mm (2"x 4") = RSI 2.3 (R13)
- 38mm x 140mm (2"x 6") = RSI 3.6 (R20)

ICYNENE LD-C-50™ provides more effective performance than the equivalent R-value of air permeable insulation materials. ICYNENE LD-C-50™ is not subject to loss of R-value due to aging, windy conditions, settling, convection or air infiltration; nor will it be prone to traditional moisture intrusion via air leakage.

A FACT SHEET with R-value data is available upon request.

#### Air Permeance/Air Barrier /Air-Seal

ICYNENE LD-C-50™ fills any shaped cavity, and adheres to most construction materials, creating assemblies with very low air permeance. Additional interior or exterior air infiltration protection is subject to applicable codes.

Air permeability of a 2" x 6" wood framed wall assembly (ASTM E2178):

- 0.01 L/s-m<sup>2</sup> @ 75 Pa for 140mm (5.5")

All buildings insulated and air sealed with ICYNENE LD-C-50™ must be designed to include adequate mechanical ventilation outdoor air supply.

See CAN/CSA F-326 - Residential Mechanical Ventilation, HRAI (Heating, Refrigeration and Air Conditioning Institute of Canada) Digest, ASHRAE (American Society of Heating, Refrigeration and Air-Conditioning Engineers) guidelines, or any other acceptable good engineering practice.

#### Water Vapour Permeance

ICYNENE LD-C-50™ is water vapour permeable and allows moisture to diffuse through the insulation and dissipate from the building envelope.

Water vapour transmission properties: (ASTM E96 Desiccant Method)

- 1218 ng / (Pa.s.m<sup>2</sup>) @ 25.4mm  
(21 Perms @ 1")

In those situations that warrant a vapour barrier, a supplemental layer of polyethylene may be used. Alternately, low vapour permeance paint either directly on the foam or as a primer for the interior drywall may be used.

#### Water Absorption Properties

Water can be forced into the foam under pressure because it is open celled. Water will drain by gravity, given favourable drying potential, and upon drying all chemical and physical properties are fully restored.

#### Acoustical Properties

Performance in a full fill 38mm x 89mm wood stud wall.

STC Sound Transmission Class - 37

|           |     |     |     |      |      |      |
|-----------|-----|-----|-----|------|------|------|
| Hz. Freq. | 125 | 250 | 500 | 1000 | 2000 | 4000 |
| ASTM E90  | 19  | 30  | 31  | 42   | 38   | 46   |

NRC Noise Reduction Coefficient - 70

|           |     |     |     |      |      |      |
|-----------|-----|-----|-----|------|------|------|
| Hz. Freq. | 125 | 250 | 500 | 1000 | 2000 | 4000 |
| ASTM C423 | .11 | .43 | .89 | .72  | .71  | .67  |

#### Burn Characteristics

ICYNENE LD-C-50™ is a combustible product and is therefore, consumed by flame, but will not sustain flame upon removal of the flame source. It leaves a charred foam residue. It will not melt or drip. ICYNENE LD-C-50™ is subject to all applicable National and/or Provincial building codes regarding fire prevention. Requirements for Thermal Barrier coverings must be met as per the applicable building code having jurisdiction.

#### CANADA Fire Testing

|                                           |     |
|-------------------------------------------|-----|
| CAN/ULC S102<br>Flame Spread/Smoke (FSCI) | 275 |
| CAN/ULC S127-04<br>Flame Spread (FSC2)    | 450 |

#### Electrical Wiring

ICYNENE LD-C-50™ has been evaluated with energized 14/3 and 12/2 residential wiring (max. 50°C/122°F). It is chemically compatible with typical electrical wiring coverings.

Note: For any insulation of older knob and tube wiring, please reference local electrical code.

#### Corrosion

ICYNENE LD-C-50™ did not cause corrosion when evaluated in contact with steel at 48°C (120°F) and 85% relative humidity conditions.

## Plastic Piping

ICYNENE LD-R-50™ is compatible in direct contact with CPVC piping systems, as per Paschal Engineering Study for the Spray Polyurethane Foam Alliance (SPFA).

## Bacterial or Fungal Growth and Food Value

Independent testing conducted by Texas Tech University has confirmed that ICYNENE LD-C-50™ is not a source of food for mould; and as an air barrier material, it resists the airborne introduction of moisture, nutrients, and mould spores into the building envelope.

## Environmental / Health / Safety

ICYNENE LD-C-50™ is 100% water-blown and therefore contains no ozone-depleting blowing agents. It is also PBDE-free. It has been thoroughly evaluated for in-situ emissions by industry and government experts. VOC emissions have been deemed safe in that they are below 1/100th of the Threshold Limit Values (TLV) within hours following the application of ICYNENE LD-C-50™.

Proper handling and use is required to avoid exposure to reactive chemicals in their unreacted state. For more information, contact the Spray Polyurethane Foam Alliance or the American Chemistry Council. Newly insulated areas have been shown to be safe for occupancy 24 hours after installation is complete.

The reaction used to create ICYNENE LD-C-50™ generates Carbon Dioxide to expand the foam. Carbon Dioxide has a very low Global Warming Potential (GWP of 1).

Not intended for exterior use. Not to be installed within 75mm (3") of heat emitting devices or where the temperature is in excess of 93°C (200°F), as per ASTM C411 or in accordance with applicable codes.

## 5. INSTALLATION

ICYNENE LD-C-50™ is installed by a network of Licensed Dealers, trained in the installation of ICYNENE LD-C-50™. Installation is generally independent of environmental conditions. It can be installed in hot, humid or freezing conditions. Surface preparation is generally not necessary. Within seconds, the foaming process is complete.

For information on Health and Safety, refer to the Spray Polyurethane Foam Alliance Health and Safety guidance documents at [www.spraypolyurethane.com](http://www.spraypolyurethane.com)

## 6. AVAILABILITY

Check regional Yellow Pages™ or contact Icynene Inc. at 800-758-7325 or our website at [www.icynene.com](http://www.icynene.com) for a local Icynene Licensed Dealer.

## 7. WARRANTY

WHEN INSTALLED PROPERLY IN ACCORDANCE WITH INSTRUCTIONS, THE COMPANY WARRANTS THAT THE PROPERTIES OF THE PRODUCT MEET PRODUCT SPECIFICATIONS AS OUTLINED IN THIS PRODUCT SPECIFICATION SHEET. SAVE AND EXCEPT ANY EXCLUSIONS REFERENCED IN THE WARRANTY.

## 8. TECHNICAL

Icynene Licensed Dealers and Icynene Inc. provide support on both technical and regulatory issues. Architectural specifications in CSI 3-Part format and design details are available upon request.

## 9. REGULATORY

ICYNENE LD-C-50™ has been independently evaluated by the Canadian Construction Materials Centre (CCMC). For regulatory issues concerning ICYNENE LD-C-50™ in Canada, please see CCMC Evaluation Report 12070-R, The NBCC (National Building Code of Canada) or applicable Provincial Building Codes.

## 10. RELATED REFERENCES

All physical properties were determined through testing by accredited third party agencies. Icynene Inc. reserves the right to change specifications in its effort of continuous improvement. Please confirm that technical data literature is current.

## 11. PACKAGING AND STORAGE

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| Packaging     | 208 L [55 U.S. gallon] steel drums                                   |
| Component 'A' | 249 kg (550 lb.) per drum<br><br>Base Seal® MDI                      |
| Component 'B' | 226 kg (500 lb.) per drum<br><br>ICYNENE LD-C-50™ (Gold Seal®) Resin |

### Storage

Component A, Base Seal®, Polymeric Isocyanate and ICYNENE LD-C-50™ Resin (Component B) ideally should be stored between 15°C (60°F) and 32°C (90°F).

Component A should be protected from freezing.

Component B can be frozen but must be protected from overheating (49°C /120°F) and prolonged storage above 38°C /100°F. Component B separates during storage and should be mixed thoroughly prior to use.

## 12. INSTALLATION SPECIFICATIONS

Must be installed by Icynene Licensed Dealers. Refer to the Icynene Installer's Manual for expanded information.



# CERTIFICATE OF CONFORMITY

This is to certify that  
**Icynene LD-C-50™**

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## Product description

Icynene LD-C-50™ is a polyurethane micro-cellular plastic foam insulation, which is sprayed onto a wall or into a wall cavity to form an insulation system.

## Product purpose or use

Thermal insulation for use in walls, ceiling or floors. (assessed based on ASTM C518, referenced in AS/NZS 4859.1:2002 including Amendment 1)

**Table: 1** - Thermal Resistance

| Thickness, in mm | Thermal Resistance, in m <sup>2</sup> .K/W | Equivalent R rating |
|------------------|--------------------------------------------|---------------------|
| 25               | 0.65                                       | 0.65                |
| 50               | 1.30                                       | 1.30                |
| 75               | 1.95                                       | 1.95                |
| 100              | 2.60                                       | 2.60                |
| 125              | 3.15                                       | 3.15                |
| 150              | 3.90                                       | 3.90                |
| 175              | 4.55                                       | 4.55                |
| 200              | 5.20                                       | 5.20                |

## Certificate holder

**Icynene Inc**  
6747 Campobello Road, Mississauga, Ontario L5N2I7, Canada Tel: + 1-905-363-4040  
Web site: [www.icynene.com](http://www.icynene.com)



[WWW.JAS-ANZ.ORG/REGISTER](http://WWW.JAS-ANZ.ORG/REGISTER)

| CodeMark Certification Body                                                                                                                         |  |  | <u>13/05/2010</u> | <u>15/08/2013</u> | <u>26/07/2016</u> | <u>GM-CM30009 Rev D</u> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------------|
| Global-Mark Pty Ltd, Suite 4.07, 32 Delhi Road, North Ryde NSW 2113, Australia - <a href="http://www.Global-Mark.com.au">www.Global-Mark.com.au</a> | Herve Michoux<br>Managing Director                                                  | Unrestricted Building<br>Certifier, Peter Gardner                                   | Date of issue     | Last update       | Date of expiry    | Certificate Number      |

This Certificate of Conformity is issued by an accredited certification body under arrangement with JAS-ANZ. The ABCB does not in any way warrant, guarantee or represent that the Product the subject of this Certificate of Conformity conforms with the BCA, nor accepts any liability arising out of the use of the Product. The ABCB disclaims to the extent permitted by law, all liability (including negligence) for claims of losses, expenses, damages and costs arising as a result of the use of the product(s) referred to in this Certificate.

It is advised to check that this Certificate of Conformity is currently valid and not withdrawn, suspended or superseded by a later issue by referring to the ABCB website, [www.abcb.gov.au](http://www.abcb.gov.au)

This is to certify that

**Icynene LD-C-50™**

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**Complies with the Building Code of Australia 2013:**

1. Volume One JP1 in respect of energy efficiency of walls, ceilings and floors, in applications where complying thermal resistances have been determined using the values in Table 1;
2. Volume Two P2.6.1 in respect of energy efficiency of walls, ceilings and floors, in applications where complying thermal resistances have been determined using the values in Table 1

**State Additions:**

Volume Two – Victoria P2.6.1.

The product will contribute to compliance when installed in accordance with:

- Volume One Clauses J1.2, (a), (c) and (d), J1.3, SA J1.3, J 1.5 and J1.6; and
- Volume Two Clauses 3.12.1.1(a) and (c), 3.12.1.2, 3.12.1.4 and 3.12.1.5.

**Subject to the following conditions and limitations:**

1. BCA Volume One J1.2(b) and BCA Volume Two 3.12.1.1 (b) do not apply.
2. The thermal resistances in Table 1 are of the Icynene semi-rigid polyurethane foam insulation alone (i.e. corresponding to “added insulation” as used in the BCA). The total thermal resistance of a wall, floor or ceiling system is the sum of values for the external air film (0.04), external cladding, air space or cavity (if applicable), Icynene insulation (from Table 1), internal cladding, and internal air space (0.12).
3. The calculated values do not account for thermal bridging at studs and the like.
4. The calculated values are based on the calculated upper 10 percentile characteristic value for thermal conductivity of 0.038 W/m.K.
6. The reported thermal resistances are based on a mean temperature of was 24 °C, and a temperature differential of 22 °C.
7. These values may be used, in conjunction with the thermal properties of other components, to satisfy:
  - Performance requirement BCA Volume One JP1, using verification method JV3.
  - Performance requirement BCA Volume Two P2.6.1 using verification method V2.6.2.2.
  - Deemed-to-Satisfy provisions BCA Volume One J1.3 and SA J1.3, using Tables J1.3a; BCA Volume One J1.5, using Tables J1.5a and J1.5b; and, BCA Volume One J1.6, using Table J1.6.
  - Deemed-to-Satisfy provisions BCA Volume Two 3.12.1.2 using Table 3.12.1.1a; BCA Volume Two 3.12.1.4, using Tables 3.12.1.3a and 3.12.1.3b; and, 3.12.1.5, using Table 3.12.1.4.
8. Product installation shall be carried out by a person trained and approved by Icynene as an Approved Installer, in accordance with the Icynene Installer Australia New Zealand review November 2009, and in accordance with AS3999:1992 (including Amendment 1).
9. This Certificate excludes compliance with:
  - BCA Volume One Section C: compliance for non-combustibility, fire hazard properties when used as a wall or ceiling lining, fire hazard properties when used as a composite member (e.g. insulation within a wall), fire hazard properties generally and regarding fire resistance or fire resistance level.
  - BCA Volume Two Part 3.7 for non-combustibility and regarding fire resistance or fire resistance level.
  - BCA Volume One Section Part G5 and BCA Volume Two P2.3.4 and P2.3.5 regarding bushfire areas.
  - Vermin proofing.
  - Condensation control.
10. Where Icynene is installed, electrical installation must comply with AS/NZS 3000:2007
11. This Certification is valid for the determination of thermal resistance of wall in accordance with BCA 2013 and in those States and Territories where there are variations to the BCA 2013

**End of document**

## ICYNENE LD-C-50 PART A

Era Polymers Pty Ltd

Chemwatch: 9-41072

Version No: 1.2

Safety Data Sheet according to WHS and ADG requirements

Print Date: 19/11/2013

Issue Date: 19/11/2013

S.GHS.AUS.EN

### SECTION 1 Identification of the substance / mixture and of the company / undertaking

| Product Identifier             |                        |
|--------------------------------|------------------------|
| Product name:                  | ICYNENE LD-C-50 PART A |
| Chemical Name:                 | Not Applicable         |
| Proper shipping name:          | Not Applicable         |
| Chemical formula:              | Not Applicable         |
| Other means of identification: |                        |
| CAS number:                    | Not Applicable         |

| Relevant identified uses of the substance or mixture and uses advised against |                                                                                                    |
|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Relevant identified uses:                                                     | Use according to manufacturer's directions.<br>Component for the manufacture of urethane polymers. |

| Details of the supplier of the safety data sheet |                                       |  |  |
|--------------------------------------------------|---------------------------------------|--|--|
| Registered company name:                         | Era Polymers Pty Ltd                  |  |  |
| Address:                                         | 25-27 Green Street 2019 NSW Australia |  |  |
| Telephone:                                       | +61 (0)2 9666 3788                    |  |  |
| Fax:                                             | +61 (0)2 9666 4805                    |  |  |
| Website:                                         | www.erapol.com.au                     |  |  |
| Email:                                           | erapol@erapol.com.au                  |  |  |

| Emergency telephone number         |           |  |  |
|------------------------------------|-----------|--|--|
| Association / Organisation:        | CHEMWATCH |  |  |
| Emergency telephone numbers:       |           |  |  |
| Other emergency telephone numbers: |           |  |  |

| CHEMWATCH EMERGENCY RESPONSE                                                            |                      |                      |
|-----------------------------------------------------------------------------------------|----------------------|----------------------|
| Primary Number                                                                          | Alternative Number 1 | Alternative Number 2 |
| 1800 039 008                                                                            | +612 9186 1132       |                      |
| Once connected and if the message is not in your preferred language then please dial 01 |                      |                      |

### SECTION 2 Hazards identification

| Classification of the substance or mixture                                                        |  |
|---------------------------------------------------------------------------------------------------|--|
| HAZARDOUS CHEMICAL. NON-DANGEROUS GOODS. According to the Model WHS Regulations and the ADG Code. |  |

|                   |    |
|-------------------|----|
| Poisons Schedule: | S6 |
|-------------------|----|

|                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GHS Classification <sup>[1]</sup> :                                                                                                                                                                                                       |
| Acute Toxicity (Inhalation) Category 4, Respiratory Sensitizer Category 1, STOT - SE (Resp. Irr.) Category 3, Carcinogen Category 2, Eye Irrit. 2, STOT - RE Category 2, Skin Corrosion/Irritation Category 2, Skin Sensitizer Category 1 |

**Legend:** 1. Classified by Chemwatch; 2. Classification drawn from HSIS; 3. Classification drawn from EC Directive 1272/2008 - Annex VI

| Label elements     |
|--------------------|
| GHS label elements |



|              |        |
|--------------|--------|
| Signal word: | DANGER |
|--------------|--------|

| Hazard statement(s): |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| H315                 | Causes skin irritation                                                    |
| H317                 | May cause an allergic skin reaction                                       |
| H319                 | Causes serious eye irritation                                             |
| H332                 | Harmful if inhaled                                                        |
| H334                 | May cause allergy or asthma symptoms or breathing difficulties if inhaled |
| H335                 | May cause respiratory irritation                                          |
| H351                 | Suspected of causing cancer                                               |

H373 May cause damage to organs through prolonged or repeated exposure

| Precautionary statement(s): Prevention |                                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| P201                                   | Obtain special instructions before use.                                                                                          |
| P202                                   | Do not handle until all safety precautions have been read and understood.                                                        |
| P260                                   | Do not breathe dust/fume/gas/mist/vapours/spray.                                                                                 |
| P261                                   | Avoid breathing dust/fume/gas/mist/vapours/spray.                                                                                |
| P264                                   | Wash all exposed external body areas thoroughly after handling.                                                                  |
| P271                                   | Use only outdoors or in a well-ventilated area.                                                                                  |
| P272                                   | Contaminated work clothing should not be allowed out of the workplace.                                                           |
| P280                                   | Wear protective gloves/protective clothing/eye protection/face protection.                                                       |
| P284                                   | [In case of inadequate ventilation] wear respiratory protection.                                                                 |
| Precautionary statement(s): Response   |                                                                                                                                  |
| P302+P352                              | IF ON SKIN: Wash with plenty of water and soap                                                                                   |
| P304+P340                              | IF INHALED: Remove person to fresh air and keep comfortable for breathing.                                                       |
| P305+P351+P338                         | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P308+P313                              | IF exposed or concerned: Get medical advice/attention.                                                                           |
| P312                                   | Call a POISON CENTER/doctor/physician/first aider/if you feel unwell.                                                            |
| P314                                   | Get medical advice/attention if you feel unwell.                                                                                 |
| P321                                   | Specific treatment (see advice on this label).                                                                                   |
| P332+P313                              | If skin irritation occurs: Get medical advice/attention.                                                                         |
| P333+P313                              | If skin irritation or rash occurs: Get medical advice/attention.                                                                 |
| P337+P313                              | If eye irritation persists: Get medical advice/attention.                                                                        |
| P342+P311                              | If experiencing respiratory symptoms: Call a POISON CENTER/doctor/physician/first aider                                          |
| P362+P364                              | Take off contaminated clothing and wash it before reuse.                                                                         |
| Precautionary statement(s): Storage    |                                                                                                                                  |
| P403+P233                              | Store in a well-ventilated place. Keep container tightly closed.                                                                 |
| P405                                   | Store locked up.                                                                                                                 |
| Precautionary statement(s): Disposal   |                                                                                                                                  |
| P501                                   | Dispose of contents/container to authorised chemical landfill or if organic to high temperature incineration                     |

### SECTION 3 Composition / information on ingredients

#### Substances

See section below for composition of Mixtures

| Mixtures  |           |                                                         |
|-----------|-----------|---------------------------------------------------------|
| CAS No    | %[weight] | Name                                                    |
| 9016-87-9 | 100       | <a href="#">POLYMERIC DIPHENYLMETHANE DIISOCYANATE</a>  |
| 101-68-8  | 40-50     | <a href="#">4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI)</a> |

Note: CAS 101-68-8 is an MDI isomer that is part of CAS 9016-87-9

### SECTION 4 First aid measures

#### Description of first aid measures

##### Eye Contact:

- If this product comes in contact with the eyes:
- Immediately hold eyelids apart and flush the eye continuously with running water.
  - Ensure complete irrigation of the eye by keeping eyelids apart and away from eye and moving the eyelids by occasionally lifting the upper and lower lids.
  - Continue flushing until advised to stop by the Poisons Information Centre or a doctor, or for at least 15 minutes.
  - Transport to hospital or doctor without delay.
  - Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.

##### Skin Contact:

- If skin contact occurs:
- Immediately remove all contaminated clothing, including footwear.
  - Flush skin and hair with running water (and soap if available).
  - Seek medical attention in event of irritation.

##### Inhalation:

- If fumes or combustion products are inhaled remove from contaminated area.
- Lay patient down. Keep warm and rested.
- Prostheses such as false teeth, which may block airway, should be removed, where possible, prior to initiating first aid procedures.
- Apply artificial respiration if not breathing, preferably with a demand valve resuscitator, bag-valve mask device, or pocket mask as trained. Perform CPR if necessary.
- Transport to hospital, or doctor, without delay.

Following uptake by inhalation, move person to an area free from risk of further exposure. Oxygen or artificial respiration should be administered as needed. Asthmatic-type symptoms may develop and may be immediate or delayed up to several hours. Treatment is essentially symptomatic. A physician should be consulted.

##### Ingestion:

- **IF SWALLOWED, REFER FOR MEDICAL ATTENTION, WHERE POSSIBLE, WITHOUT DELAY.**
- For advice, contact a Poisons Information Centre or a doctor.
- Urgent hospital treatment is likely to be needed.
- In the mean time, qualified first-aid personnel should treat the patient following observation and employing supportive measures as indicated by the patient's condition.
- If the services of a medical officer or medical doctor are readily available, the patient should be placed in his/her care and a copy of the MSDS should be provided. Further action will be the responsibility of the medical specialist.
- If medical attention is not available on the worksite or surroundings send the patient to a hospital together with a copy of the MSDS.

Where medical attention is not immediately available or where the patient is more than 15 minutes from a hospital or unless instructed otherwise:

- **INDUCE** vomiting with fingers down the back of the throat, **ONLY IF CONSCIOUS**. Lean patient forward or place on left side (head-down position, if possible) to maintain open airway and prevent aspiration.

**NOTE:** Wear a protective glove when inducing vomiting by mechanical means.

#### Indication of any immediate medical attention and special treatment needed

For sub-chronic and chronic exposures to isocyanates:

- This material may be a potent pulmonary sensitiser which causes bronchospasm even in patients without prior airway hyperreactivity.
- Clinical symptoms of exposure involve mucosal irritation of respiratory and gastrointestinal tracts.
- Conjunctival irritation, skin inflammation (erythema, pain vesiculation) and gastrointestinal disturbances occur soon after exposure.
- Pulmonary symptoms include cough, burning, substernal pain and dyspnoea.
- Some cross-sensitivity occurs between different isocyanates.
- Noncardiogenic pulmonary oedema and bronchospasm are the most serious consequences of exposure. Markedly symptomatic patients should receive oxygen, ventilatory support and an intravenous line.
- Treatment for asthma includes inhaled sympathomimetics (epinephrine [adrenalin], terbutaline) and steroids.
- Activated charcoal (1 g/kg) and a cathartic (sorbitol, magnesium citrate) may be useful for ingestion.
- Mydriatics, systemic analgesics and topical antibiotics (Sulamyd) may be used for corneal abrasions.
- There is no effective therapy for sensitised workers.

[Ellenhorn and Barceloux; Medical Toxicology]

**NOTE:** Isocyanates cause airway restriction in naive individuals with the degree of response dependant on the concentration and duration of exposure. They induce smooth muscle contraction which leads to bronchoconstrictive episodes. Acute changes in lung function, such as decreased FEV1, may not represent sensitivity.

[Karol & Jin, Frontiers in Molecular Toxicology, pp 56-61, 1992]

Personnel who work with isocyanates, isocyanate prepolymers or polyisocyanates should have a pre-placement medical examination and periodic examinations thereafter, including a pulmonary function test. Anyone with a medical history of chronic respiratory disease, asthmatic or bronchial attacks, indications of allergic responses, recurrent eczema or sensitisation conditions of the skin should not handle or work with isocyanates. Anyone who develops chronic respiratory distress when working with isocyanates should be removed from exposure and examined by a physician. Further exposure must be avoided if a sensitivity to isocyanates or polyisocyanates has developed.

## SECTION 5 Firefighting measures

### Extinguishing media

- Small quantities of water in contact with hot liquid may react violently with generation of a large volume of rapidly expanding hot sticky semi-solid foam.
- Presents additional hazard when fire fighting in a confined space.
- Cooling with flooding quantities of water reduces this risk.
- Water spray or fog may cause frothing and should be used in large quantities.
- Foam.

### Special hazards arising from the substrate or mixture

#### Fire Incompatibility:

- Avoid contamination with oxidising agents i.e. nitrates, oxidising acids, chlorine bleaches, pool chlorine etc. as ignition may result

### Advice for firefighters

#### Fire Fighting:

- Alert Fire Brigade and tell them location and nature of hazard.
- Wear full body protective clothing with breathing apparatus.
- Prevent, by any means available, spillage from entering drains or water course.
- Use water delivered as a fine spray to control fire and cool adjacent area.
- Avoid spraying water onto liquid pools.

#### Fire/Explosion Hazard:

- Combustible.
- Moderate fire hazard when exposed to heat or flame.
- When heated to high temperatures decomposes rapidly generating vapour which pressures and may then rupture containers with release of flammable and highly toxic isocyanate vapour.
- Burns with acrid black smoke and poisonous fumes.
- Combustion yields traces of highly toxic hydrogen cyanide HCN, plus toxic nitrogen oxides NOx and carbon monoxide.

## SECTION 6 Accidental release measures

### Personal precautions, protective equipment and emergency procedures

#### Minor Spills:

- Remove all ignition sources.
- Clean up all spills immediately.
- Avoid breathing vapours and contact with skin and eyes.
- Control personal contact with the substance, by using protective equipment.
- Contain and absorb spill with sand, earth, inert material or vermiculite.

#### Major Spills:

- Liquid Isocyanates and high isocyanate vapour concentrations will penetrate seals on self contained breathing apparatus - SCBA should be used inside encapsulating suit where this exposure may occur.

For isocyanate spills of less than 40 litres (2 m2):

- Evacuate area from everybody not dealing with the emergency, keep them upwind and prevent further access, remove ignition sources and, if inside building, ventilate area as well as possible.
- Notify supervision and others as necessary.
- Put on personal protective equipment (suitable respiratory protection, face and eye protection, protective suit, gloves and impermeable boots).

Personal Protective Equipment advice is contained in Section 8 of the MSDS.

## SECTION 7 Handling and storage

### Precautions for safe handling

#### Safe handling

- **DO NOT allow clothing wet with material to stay in contact with skin**
- Avoid all personal contact, including inhalation.
- Wear protective clothing when risk of exposure occurs.
- Use in a well-ventilated area.

#### Other information

for commercial quantities of isocyanates:

- Isocyanates should be stored in adequately banded areas. Nothing else should be kept within the same bunding. Pre-polymers need not be segregated. Drums of isocyanates should be stored under cover, out of direct sunlight, protected from rain, protected from physical damage and well away from moisture, acids and alkalis.

### Conditions for safe storage, including any incompatibilities

#### Suitable container:

- Metal can or drum
- Packaging as recommended by manufacturer.
- Check all containers are clearly labelled and free from leaks.

#### Storage incompatibility:

- Avoid cross contamination between the two liquid parts of product (kit).
- If two part products are mixed or allowed to mix in proportions other than manufacturer's recommendation, polymerisation with gelation and evolution of heat (exotherm) may occur.
- This excess heat may generate toxic vapour
- Avoid any contamination of this material as it is very reactive and any contamination is potentially hazardous
- Avoid reaction with water, alcohols and detergent solutions.

#### Package Material Incompatibilities:

## SECTION 8 Exposure controls / personal protection

### Control parameters

#### Occupational Exposure Limits (OEL)

#### INGREDIENT DATA

| Source                       | Ingredient                              | Material name             | TWA         | STEL        | Peak | Notes |
|------------------------------|-----------------------------------------|---------------------------|-------------|-------------|------|-------|
| Australia Exposure Standards | POLYMERIC DIPHENYLMETHANE DIISOCYANATE  | Isocyanates, all (as-NCO) | 0.02 (mgm3) | 0.07 (mgm3) |      |       |
| Australia Exposure Standards | 4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI) | Isocyanates, all (as-NCO) | 0.02 (mgm3) | 0.07 (mgm3) |      |       |

#### Emergency Limits

| Ingredient                              | TEEL-0        | TEEL-1        | TEEL-2      | TEEL-3        |
|-----------------------------------------|---------------|---------------|-------------|---------------|
| POLYMERIC DIPHENYLMETHANE DIISOCYANATE  | 2(ppm)        | 6(ppm)        | 40(ppm)     | 200(ppm)      |
| 4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI) | 0.05 / 5(ppm) | 0.2 / 15(ppm) | 2 / 25(ppm) | 25 / 125(ppm) |

| Ingredient                              | Original IDLH | Revised IDLH |
|-----------------------------------------|---------------|--------------|
| 4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI) | 100(mgm3)     | 75(mgm3)     |

### Exposure controls

#### Appropriate engineering controls

- All processes in which isocyanates are used should be enclosed wherever possible.
- Total enclosure, accompanied by good general ventilation, should be used to keep atmospheric concentrations below the relevant exposure standards.
- If total enclosure of the process is not feasible, local exhaust ventilation may be necessary. Local exhaust ventilation is essential where lower molecular weight isocyanates (such as TDI or HDI) is used or where isocyanate or polyurethane is sprayed.
- Where other isocyanates or pre-polymers are used and aerosol formation cannot occur, local exhaust ventilation may not be necessary if the atmospheric concentration can be kept below the relevant exposure standards.

#### Personal protection



#### Eye and face protection:

- Safety glasses with side shields.
- Chemical goggles.
- Contact lenses may pose a special hazard; soft contact lenses may absorb and concentrate irritants. A written policy document, describing the wearing of lens or restrictions on use, should be created for each workplace or task. This should include a review of lens absorption and adsorption for the class of chemicals in use and an account of injury experience.

#### Skin protection:

See Hand protection below

#### Hand protection:

##### NOTE:

- The material may produce skin sensitisation in predisposed individuals. Care must be taken, when removing gloves and other protective equipment, to avoid all possible skin contact.
- Contaminated leather items, such as shoes, belts and watch-bands should be removed and destroyed.

The selection of suitable gloves does not only depend on the material, but also on further marks of quality which vary from manufacturer to manufacturer.

#### Body protection:

See Other protection below

#### Other protection:

All employees working with isocyanates must be informed of the hazards from exposure to the contaminant and the precautions necessary to prevent damage to their health. They should be made aware of the need to carry out their work so that as little contamination as possible is produced, and of the importance of the proper use of all safeguards against exposure to themselves and their fellow workers. Adequate training, both in the proper execution of the task and in the use of all associated engineering controls, as well as of any personal protective equipment, is essential. Employees exposed to contamination hazards should be educated in the need for, and proper use of, facilities, clothing and equipment and thereby maintain a high standard of personal cleanliness. Special attention should be given to ensuring that all personnel understand instructions, especially newly recruited employees and those with local-language difficulties, where they are known.

#### Thermal hazards:

#### Recommended material(s):

##### GLOVE SELECTION INDEX

Glove selection is based on a modified presentation of the:

**"Forsberg Clothing Performance Index".**

The effect(s) of the following substance(s) are taken into account in the

**computer-generated**

ICYNENE LD-C-50 PART A

| Material | CPI |
|----------|-----|
|----------|-----|

\* CPI - Chemwatch Performance Index

A: Best Selection

B: Satisfactory; may degrade after 4 hours continuous immersion

C: Poor to Dangerous Choice for other than short term immersion

**NOTE**

#### Respiratory protection:

Where the concentration of gas/particulates in the breathing zone, approaches or exceeds the "Exposure Standard" (or ES), respiratory protection is required.

Degree of protection varies with both face-piece and Class of filter; the nature of protection varies with Type of filter.

| Required Minimum Protection Factor | Half-Face Respirator | Full-Face Respirator | Powered Air Respirator |
|------------------------------------|----------------------|----------------------|------------------------|
| up to 10 x ES                      | A-AUS                | -                    | A-PAPR-AUS / Class 1   |
| up to 50 x ES                      | -                    | A-AUS / Class 1      | -                      |
| up to 100 x ES                     | -                    | A-2                  | A-PAPR-2 ^             |

^ - Full-face

A(All classes) = Organic vapours, B AUS or B1 = Acid gasses, B2 = Acid gas or hydrogen cyanide(HCN), B3 = Acid gas or hydrogen cyanide(HCN), E = Sulfur dioxide(SO2), G = Agricultural chemicals, K = Ammonia(NH3), Hg = Mercury, NO = Oxides of nitrogen, MB = Methyl bromide, AX = Low boiling point organic compounds(below 65 degC)

## SECTION 9 Physical and chemical properties

### Information on basic physical and chemical properties

|                                              |                                      |                                         |                |
|----------------------------------------------|--------------------------------------|-----------------------------------------|----------------|
| Appearance                                   | Brown liquid with slight musty odour |                                         |                |
| Physical state                               | Liquid                               | Relative density (Water = 1)            | 1.24           |
| Odour                                        | Slight                               | Partition coefficient n-octanol / water |                |
| Odour threshold                              |                                      | Auto-ignition temperature (°C)          | >600           |
| pH (as supplied)                             | Not Applicable                       | Decomposition temperature               | >230           |
| Melting point / freezing point (°C)          | Not Applicable                       | Viscosity (cSt)                         |                |
| Initial boiling point and boiling range (°C) |                                      | Molecular weight (g/mol)                | Not Applicable |
| Flash point (°C)                             | >204                                 | Taste                                   |                |
| Evaporation rate                             |                                      | Explosive properties                    |                |
| Flammability                                 |                                      | Oxidising properties                    |                |
| Upper Explosive Limit (%)                    |                                      | Surface Tension (dyn/cm or mN/m)        |                |
| Lower Explosive Limit (%)                    |                                      | Volatile Component (%vol)               |                |
| Vapour pressure (kPa)                        |                                      | Gas group                               |                |
| Solubility in water (g/L)                    | Reacts                               | pH as a solution(1%)                    | Not Applicable |
| Vapour density (Air = 1)                     | 8.5                                  |                                         |                |

## SECTION 10 Stability and reactivity

|                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactivity:                                                                                                                                                                                                                |
| See section 7                                                                                                                                                                                                              |
| Chemical stability:                                                                                                                                                                                                        |
| <ul style="list-style-type: none"> <li>Presence of incompatible materials.</li> <li>Product is considered stable.</li> <li>Hazardous polymerisation will not occur.</li> <li>Presence of elevated temperatures.</li> </ul> |
| Possibility of hazardous reactions:                                                                                                                                                                                        |
| See section 7                                                                                                                                                                                                              |
| Conditions to avoid:                                                                                                                                                                                                       |
| See section 7                                                                                                                                                                                                              |
| Incompatible materials:                                                                                                                                                                                                    |
| See section 7                                                                                                                                                                                                              |
| Hazardous decomposition products:                                                                                                                                                                                          |
| See section 5                                                                                                                                                                                                              |

## SECTION 11 Toxicological information

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Information on toxicological effects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Inhaled:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Evidence shows, or practical experience predicts, that the material produces irritation of the respiratory system, in a substantial number of individuals, following inhalation. In contrast to most organs, the lung is able to respond to a chemical insult by first removing or neutralising the irritant and then repairing the damage. The repair process, which initially evolved to protect mammalian lungs from foreign matter and antigens, may however, produce further lung damage resulting in the impairment of gas exchange, the primary function of the lungs. Respiratory tract irritation often results in an inflammatory response involving the recruitment and activation of many cell types, mainly derived from the vascular system. The vapour/mist may be highly irritating to the upper respiratory tract and lungs; the response may be severe enough to produce bronchitis and pulmonary oedema.                      |
| Ingestion:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Accidental ingestion of the material may be seriously damaging to the health of the individual; animal experiments indicate that ingestion of less than 40 gram may be fatal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Skin Contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Evidence exists, or practical experience predicts, that the material either produces inflammation of the skin in a substantial number of individuals following direct contact, and/or produces significant inflammation when applied to the healthy intact skin of animals, for up to four hours, such inflammation being present twenty-four hours or more after the end of the exposure period. Skin irritation may also be present after prolonged or repeated exposure; this may result in a form of contact dermatitis (nonallergic). The dermatitis is often characterised by skin redness (erythema) and swelling (oedema) which may progress to blistering (vesiculation), scaling and thickening of the epidermis. At the microscopic level there may be intercellular oedema of the spongy layer of the skin (spongiosis) and intracellular oedema of the epidermis. The material may accentuate any pre-existing dermatitis condition |
| Eye:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Evidence exists, or practical experience predicts, that the material may cause eye irritation in a substantial number of individuals and/or may produce significant ocular lesions which are present twenty-four hours or more after instillation into the eye(s) of experimental animals. Repeated or prolonged eye contact may cause inflammation characterised by a temporary redness (similar to windburn) of the conjunctiva (conjunctivitis); temporary impairment of vision and/or other transient eye damage/ulceration may occur.                                                                                                                                                                                                                                                                                                                                                                                                       |
| Chronic:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| On the basis, primarily, of animal experiments, concern has been expressed that the material may produce carcinogenic or mutagenic effects; in respect of the available information, however, there presently exists inadequate data for making a satisfactory assessment. Long-term exposure to respiratory irritants may result in disease of the airways involving difficult breathing and related systemic problems. Practical evidence shows that inhalation of the material is capable of inducing a sensitisation reaction in a substantial number of individuals at a greater frequency than would be expected from the response of a normal population. Pulmonary sensitisation, resulting in hyperactive airway dysfunction and pulmonary allergy may be accompanied by fatigue, malaise and aching. Significant symptoms of exposure may persist for extended periods, even after exposure ceases.                                    |

| TOXICITY                                                                                                | IRRITATION                       |
|---------------------------------------------------------------------------------------------------------|----------------------------------|
| ICYNENE LD-C-50 PART A                                                                                  |                                  |
| POLYMERIC DIPHENYLMETHANE DIISOCYANATE                                                                  |                                  |
| Dermal (rabbit) LD50: >9400 mg/kg                                                                       | Eye (rabbit): 100 mg - mild      |
| Inhalation (rat) LC50: 490 mg/m3/4h                                                                     |                                  |
| Oral (rat) LD50: 43000 mg/kg                                                                            |                                  |
| 4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI)                                                                 |                                  |
| Dermal (rabbit) LD50: >6200 mg/kg *                                                                     | [* = Bayer CCINFO 2133615]       |
| Inhalation (Rat) LC50: 178 mg/m3                                                                        | Dermal Sensitiser *              |
| Oral (mouse) LD50: 2200 mg/kg                                                                           | Respiratory Sensitiser (g.pig) * |
| Oral (Rat) LD50: 9200 mg/kg                                                                             | Skin (rabbit): 500 mg /24 hours  |
| ICYNENE LD-C-50 PART A, POLYMERIC DIPHENYLMETHANE DIISOCYANATE, 4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI) |                                  |

The following information refers to contact allergens as a group and may not be specific to this product.

Contact allergies quickly manifest themselves as contact eczema, more rarely as urticaria or Quincke's oedema. The pathogenesis of contact eczema involves a cell-mediated (T lymphocytes) immune reaction of the delayed type. Other allergic skin reactions, e.g. contact urticaria, involve antibody-mediated immune reactions. The significance of the contact allergen is not simply determined by its sensitisation potential: the distribution of the substance and the opportunities for contact with it are equally important.

|                                    |                                                                 |                           |                                   |
|------------------------------------|-----------------------------------------------------------------|---------------------------|-----------------------------------|
| Acute Toxicity:                    | Acute Toxicity (Inhalation) Category 4                          | Carcinogenicity:          | Carcinogen Category 2             |
| Skin Irritation/Corrosion:         | Skin Corrosion/Irritation Category 2                            | Reproductivity:           | Not Applicable                    |
| Serious Eye Damage/Irritation:     | Eye Irrit. 2                                                    | STOT - Single Exposure:   | STOT - SE (Resp. Irr.) Category 3 |
| Respiratory or Skin sensitisation: | Respiratory Sensitizer Category 1<br>Skin Sensitizer Category 1 | STOT - Repeated Exposure: | STOT - RE Category 2              |
| Mutagenicity:                      | Not Applicable                                                  | Aspiration Hazard:        | Not Applicable                    |

CMR STATUS

|                                         |                                            |         |
|-----------------------------------------|--------------------------------------------|---------|
| CARCINOGEN                              |                                            |         |
| 4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI) | Australia Exposure Standards - Carcinogens | Carc. 2 |

SECTION 12 Ecological information

Toxicity

Hydrolysis would represents the primary fate mechanism for the majority of the commercial isocyanate monomers, but, is tempered somewhat by the lack of water solubility. In the absence of hydrolysis, sorption to solids (e.g., sludge and sediments) will be the primary mechanism of removal. Biodegradation is minimal for most compounds and volatilisation is negligible. Atmospheric degradation is not expected with removal from air occurring by washout or dry deposition. Volatilisation from surface waters (e.g., lakes and rivers) is expected to take years.

Persistence and degradability

|            |                         |                  |
|------------|-------------------------|------------------|
| Ingredient | Persistence: Water/Soil | Persistence: Air |
|------------|-------------------------|------------------|

Bioaccumulative potential

|            |                 |
|------------|-----------------|
| Ingredient | Bioaccumulation |
|------------|-----------------|

Mobility in soil

|            |          |
|------------|----------|
| Ingredient | Mobility |
|------------|----------|

SECTION 13 Disposal considerations

Waste treatment methods

Product / Packaging disposal:

- Containers may still present a chemical hazard/ danger when empty.
- Return to supplier for reuse/ recycling if possible.

Otherwise:

- If container can not be cleaned sufficiently well to ensure that residuals do not remain or if the container cannot be used to store the same product, then puncture containers, to prevent re-use, and bury at an authorised landfill.
- Where possible retain label warnings and MSDS and observe all notices pertaining to the product.

SECTION 14 Transport information

Labels Required:

Marine Pollutant: NO

HAZCHEM:

Land transport (ADG): NOT REGULATED FOR TRANSPORT OF DANGEROUS GOODS

Air transport (ICAO-IATA / DGR): NOT REGULATED FOR TRANSPORT OF DANGEROUS GOODS

Sea transport (IMDG-Code / GGVSee): NOT REGULATED FOR TRANSPORT OF DANGEROUS GOODS

SECTION 15 Regulatory information

Safety, health and environmental regulations / legislation specific for the substance or mixture

POLYMERIC DIPHENYLMETHANE DIISOCYANATE(9016-87-9) is found on the following regulatory lists

"OECD List of High Production Volume (HPV) Chemicals","Australia High Volume Industrial Chemical List (HVICL)","Australia Inventory of Chemical Substances (AICS)","International Agency for Research on Cancer (IARC) - Agents Reviewed by the IARC Monographs","GESAMP/EHS Composite List - GESAMP Hazard Profiles","IMO IBC Code Chapter 17: Summary of minimum requirements","IMO MARPOL 73/78 (Annex II) - List of Noxious Liquid Substances Carried in Bulk","Australia - Victoria Occupational Health and Safety Regulations - Schedule 9: Materials at Major Hazard Facilities (And Their Threshold Quantity) Table 2","Australia National Pollutant Inventory","Australia - New South Wales Hazardous Substances Requiring Health Surveillance","Australia - Tasmania Hazardous Substances Requiring Health Surveillance","Australia Hazardous Substances Information System - Consolidated Lists","Australia Occupational Health and Safety (Commonwealth Employment) (National Standards) Regulations 1994 - Hazardous Substances Requiring Health Surveillance","Australia - Western Australia Hazardous Substances Requiring Health Surveillance","Australia Hazardous Substances Requiring Health Surveillance","Australia - Queensland Work Health and Safety Regulation - Hazardous chemicals (other than lead) requiring health monitoring","Australia Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP) - Schedule 6","Australia Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP) - Appendix E (Part 2)","Australia Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP) - Appendix F (Part 3)","Australia - South Australia - Work Health and Safety Regulations 2012 - Requirements for health monitoring - Hazardous chemicals (other than lead) requiring health monitoring","Australia - New South Wales - Work Health and Safety Regulation 2011 - Requirements for health monitoring -Hazardous chemicals (other than lead) requiring health monitoring","Australia - Tasmania - Work Health and Safety Regulations 2012 - Requirements for Health Monitoring - Hazardous chemicals (other than lead) requiring health monitoring","Australia - Northern Territories Work Health and Safety National Uniform Legislation Regulations- Requirements for health monitoring - Hazardous chemicals (other than lead) requiring health monitoring","Australia Exposure Standards","Australia Work Health and Safety Regulations 2011 - Hazardous chemicals (other than lead) requiring health monitoring"

4,4'-DIPHENYLMETHANE DIISOCYANATE (MDI)(101-68-8) is found on the following regulatory lists

"Australia National Pollutant Inventory","Australia Hazardous Substances Information System - Consolidated Lists","OECD List of High Production Volume (HPV) Chemicals","Australia High Volume Industrial Chemical List (HVICL)","Australia Inventory of Chemical Substances (AICS)","International Agency for Research on Cancer (IARC) - Agents Reviewed by the IARC Monographs","GESAMP/EHS Composite List - GESAMP Hazard Profiles","FisherTransport Information","Sigma-AldrichTransport Information","IMO IBC Code Chapter 17: Summary of minimum requirements","International Air Transport Association (IATA) Dangerous Goods Regulations","IMO MARPOL 73/78 (Annex II) - List of Noxious Liquid Substances Carried in Bulk","Australia - Victoria Occupational Health and Safety Regulations - Schedule 9: Materials at Major Hazard Facilities (And Their Threshold Quantity) Table 2","Australia - New South Wales Hazardous Substances Requiring Health Surveillance","Australia - Tasmania Hazardous Substances Requiring Health Surveillance","Australia Occupational Health and Safety (Commonwealth Employment) (National Standards) Regulations 1994 - Hazardous Substances Requiring Health Surveillance","Australia - Western Australia Hazardous Substances Requiring Health Surveillance","Australia - Queensland Work Health and Safety Regulation - Hazardous chemicals (other than lead) requiring health monitoring","Australia Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP) - Schedule 6","Australia Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP) - Appendix E (Part 2)","Australia Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP) - Appendix F (Part 3)","Australia - South Australia - Work Health and Safety Regulations 2012 - Requirements for health monitoring - Hazardous chemicals (other than lead) requiring health monitoring","Australia - New South Wales - Work Health and Safety Regulation 2011 - Requirements for health monitoring -Hazardous chemicals (other than lead) requiring health monitoring","Australia - Tasmania - Work Health and Safety Regulations 2012 - Requirements for Health Monitoring - Hazardous chemicals (other than lead) requiring health monitoring","Australia - Northern Territories Work Health and Safety National Uniform Legislation Regulations- Requirements for health monitoring - Hazardous chemicals (other than lead) requiring health monitoring","Australia Exposure Standards","Australia Work Health and Safety Regulations 2011 - Hazardous chemicals (other than lead) requiring health monitoring"

## SECTION 16 Other information

### Other information

Classification of the preparation and its individual components has drawn on official and authoritative sources as well as independent review by the Chemwatch Classification committee using available literature references.

A list of reference resources used to assist the committee may be found at:

[www.chemwatch.net/references](http://www.chemwatch.net/references)

The (M)SDS is a Hazard Communication tool and should be used to assist in the Risk Assessment. Many factors determine whether the reported Hazards are Risks in the workplace or other settings. Risks may be determined by reference to Exposures Scenarios. Scale of use, frequency of use and current or available engineering controls must be considered.

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## ICYNENE LD-C-50 PART B

Era Polymers Pty Ltd

Chemwatch: 9-41091

Version No: 1.6

Safety Data Sheet according to WHS and ADG requirements

Print Date: 19/11/2013

Issue Date: 19/11/2013

S.GHS.AUS.EN

### SECTION 1 Identification of the substance / mixture and of the company / undertaking

#### Product Identifier

|                                |                        |
|--------------------------------|------------------------|
| Product name:                  | ICYNENE LD-C-50 PART B |
| Chemical Name:                 | Not Applicable         |
| Proper shipping name:          | Not Applicable         |
| Chemical formula:              | Not Applicable         |
| Other means of identification: | Not Available          |
| CAS number:                    | Not Applicable         |

#### Relevant identified uses of the substance or mixture and uses advised against

|                           |                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relevant identified uses: | Use according to manufacturer's directions.<br>Blend of polymeric resins and additives that react with polymeric MDI to create LD-C-50, a Semi-rigid open cell polyurethane foam insulation. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Details of the supplier of the safety data sheet

|                          |                                       |
|--------------------------|---------------------------------------|
| Registered company name: | Era Polymers Pty Ltd                  |
| Address:                 | 25-27 Green Street 2019 NSW Australia |
| Telephone:               | +61 (0)2 9666 3788                    |
| Fax:                     | +61 (0)2 9666 4805                    |
| Website:                 | www.erapol.com.au                     |
| Email:                   | erapol@erapol.com.au                  |

#### Emergency telephone number

|                                    |               |
|------------------------------------|---------------|
| Association / Organisation:        | CHEMWATCH     |
| Emergency telephone numbers:       | Not Available |
| Other emergency telephone numbers: | Not Available |

#### CHEMWATCH EMERGENCY RESPONSE

| Primary Number                                                                          | Alternative Number 1 | Alternative Number 2 |
|-----------------------------------------------------------------------------------------|----------------------|----------------------|
| 1800 039 008                                                                            | +612 9186 1132       | Not Available        |
| Once connected and if the message is not in your preferred language then please dial 01 |                      |                      |

### SECTION 2 Hazards identification

#### Classification of the substance or mixture

**HAZARDOUS CHEMICAL. NON-DANGEROUS GOODS.** According to the Model WHS Regulations and the ADG Code.

|                   |    |
|-------------------|----|
| Poisons Schedule: | S5 |
|-------------------|----|

#### GHS Classification<sup>[1]</sup>:

Eye Irrit. 2, Skin Corrosion/Irritation Category 2

**Legend:**1. Classified by Chemwatch; 2. Classification drawn from HSIS; 3. Classification drawn from EC Directive 1272/2008 - Annex VI

#### Label elements

##### GHS label elements



|              |                |
|--------------|----------------|
| Signal word: | <b>WARNING</b> |
|--------------|----------------|

#### Hazard statement(s):

|      |                               |
|------|-------------------------------|
| H315 | Causes skin irritation        |
| H319 | Causes serious eye irritation |

#### Precautionary statement(s): Prevention

|      |                                                                            |
|------|----------------------------------------------------------------------------|
| P264 | Wash all exposed external body areas thoroughly after handling.            |
| P280 | Wear protective gloves/protective clothing/eye protection/face protection. |

#### Precautionary statement(s): Response

|           |                                                |
|-----------|------------------------------------------------|
| P302+P352 | IF ON SKIN: Wash with plenty of water and soap |
|-----------|------------------------------------------------|

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P321 Specific treatment (see advice on this label).

P332+P313 If skin irritation occurs: Get medical advice/attention.

P337+P313 If eye irritation persists: Get medical advice/attention.

P362+P364 Take off contaminated clothing and wash it before reuse.

#### Precautionary statement(s): Storage

Not Applicable

#### Precautionary statement(s): Disposal

Not Applicable

## SECTION 3 Composition / information on ingredients

### Substances

See section below for composition of Mixtures

### Mixtures

| CAS No        | %[weight] | Name         |
|---------------|-----------|--------------|
| Not Available | <30       | Alkanolamine |
| Not Available | <30       | Alkylamine   |

Hazardous Material Information Review Commission Registry Number No. 6626. Date of decision granting exemption: May 15, 2006.

All other substances non hazardous

Balance

## SECTION 4 First aid measures

### Description of first aid measures

#### Eye Contact:

If this product comes in contact with the eyes:

- Wash out immediately with fresh running water.
- Ensure complete irrigation of the eye by keeping eyelids apart and away from eye and moving the eyelids by occasionally lifting the upper and lower lids.
- Seek medical attention without delay; if pain persists or recurs seek medical attention.
- Removal of contact lenses after an eye injury should only be undertaken by skilled personnel.

#### Skin Contact:

If skin or hair contact occurs:

- Flush skin and hair with running water (and soap if available).
- Seek medical attention in event of irritation.

#### Inhalation:

- If fumes, aerosols or combustion products are inhaled remove from contaminated area.
- Other measures are usually unnecessary.

#### Ingestion:

- If swallowed do **NOT** induce vomiting.
- If vomiting occurs, lean patient forward or place on left side (head-down position, if possible) to maintain open airway and prevent aspiration.
- Observe the patient carefully.
- Never give liquid to a person showing signs of being sleepy or with reduced awareness; i.e. becoming unconscious.
- Give water to rinse out mouth, then provide liquid slowly and as much as casualty can comfortably drink.
- Seek medical advice.

### Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

## SECTION 5 Firefighting measures

### Extinguishing media

- Water spray or fog.
- Foam.
- Dry chemical powder.
- BCF (where regulations permit).
- Carbon dioxide.

### Special hazards arising from the substrate or mixture

#### Fire Incompatibility:

- Avoid contamination with oxidising agents i.e. nitrates, oxidising acids, chlorine bleaches, pool chlorine etc. as ignition may result

### Advice for firefighters

#### Fire Fighting:

- Alert Fire Brigade and tell them location and nature of hazard.
- Wear full body protective clothing with breathing apparatus.
- Prevent, by any means available, spillage from entering drains or water course.
- Use water delivered as a fine spray to control fire and cool adjacent area.
- Avoid spraying water onto liquid pools.

#### Fire/Explosion Hazard:

- Combustible.
- Slight fire hazard when exposed to heat or flame.
- Heating may cause expansion or decomposition leading to violent rupture of containers.
- On combustion, may emit toxic fumes of carbon monoxide (CO).
- May emit acid smoke.

## SECTION 6 Accidental release measures

### Personal precautions, protective equipment and emergency procedures

#### Minor Spills:

- Remove all ignition sources.
- Clean up all spills immediately.
- Avoid breathing vapours and contact with skin and eyes.
- Control personal contact with the substance, by using protective equipment.
- Contain and absorb spill with sand, earth, inert material or vermiculite.

#### Major Spills:

Moderate hazard.

- Clear area of personnel and move upwind.
- Alert Fire Brigade and tell them location and nature of hazard.
- Wear breathing apparatus plus protective gloves.
- Prevent, by any means available, spillage from entering drains or water course.

Personal Protective Equipment advice is contained in Section 8 of the MSDS.

## SECTION 7 Handling and storage

### Precautions for safe handling

#### Safe handling

- Overheating of ethoxylates/ alkoxyates in air should be avoided. When some ethoxylates are heated vigorously in the presence of air or oxygen, at temperatures exceeding 160 C, they may undergo exothermic oxidative degeneration resulting in self-heating and autoignition.
- Nitrogen blanketing will minimise the potential for ethoxylate oxidation. Prolonged storage in the presence of air or oxygen may cause product degradation. Oxidation is not expected when stored under a nitrogen atmosphere.

#### Other information

Ethoxylates/ alkoxyates react slowly with air or oxygen. Storage under heated conditions in the presence of air or oxygen increases reaction rate. For example, after storing at 95 F/ 35 C for 30 days in the presence of air, there is measurable oxidation of the ethoxylate. Lower temperatures will allow for longer storage time and higher temperatures will shorten the storage time if stored under an air or oxygen atmosphere.

- Store in original containers.

### Conditions for safe storage, including any incompatibilities

#### Suitable container:

For ethoxylates suitable containers include carbon steel coated with baked phenolic.

Any moisture may cause rusting of carbon steel.

If product is moisture free, uncoated carbon steel tanks may be used.

- Metal can or drum
- Packaging as recommended by manufacturer.

#### Storage incompatibility:

- Avoid reaction with oxidising agents

#### Package Material Incompatibilities:

## SECTION 8 Exposure controls / personal protection

### Control parameters

#### Occupational Exposure Limits (OEL)

#### INGREDIENT DATA

Not Available

#### Emergency Limits

| Ingredient             | TEEL-0        | TEEL-1        | TEEL-2        | TEEL-3        |
|------------------------|---------------|---------------|---------------|---------------|
| ICYNENE LD-C-50 PART B | Not Available | Not Available | Not Available | Not Available |

| Ingredient             | Original IDLH | Revised IDLH  |
|------------------------|---------------|---------------|
| ICYNENE LD-C-50 PART B | Not Available | Not Available |

### Exposure controls

#### Appropriate engineering controls

Engineering controls are used to remove a hazard or place a barrier between the worker and the hazard. Well-designed engineering controls can be highly effective in protecting workers and will typically be independent of worker interactions to provide this high level of protection.

The basic types of engineering controls are:

Process controls which involve changing the way a job activity or process is done to reduce the risk.

Enclosure and/or isolation of emission source which keeps a selected hazard "physically" away from the worker and ventilation that strategically "adds" and "removes" air in the work environment.

#### Personal protection



#### Eye and face protection:

- Safety glasses with side shields
- Chemical goggles.
- Contact lenses may pose a special hazard; soft contact lenses may absorb and concentrate irritants. A written policy document, describing the wearing of lens or restrictions on use, should be created for each workplace or task. This should include a review of lens absorption and adsorption for the class of chemicals in use and an account of injury experience.

#### Skin protection:

See Hand protection below

#### Hand protection:

The selection of suitable gloves does not only depend on the material, but also on further marks of quality which vary from manufacturer to manufacturer. Where the chemical is a preparation of several substances, the resistance of the glove material can not be calculated in advance and has therefore to be checked prior to the application.

The exact break through time for substances has to be obtained from the manufacturer of the protective gloves and has to be observed when making a final choice.

Suitability and durability of glove type is dependent on usage.

#### Body protection:

See Other protection below

Other protection:

- Overalls.
- P.V.C. apron.
- Barrier cream.
- Skin cleansing cream.

Thermal hazards:

Recommended material(s):

GLOVE SELECTION INDEX

Glove selection is based on a modified presentation of the:  
"Forsberg Clothing Performance Index".  
The effect(s) of the following substance(s) are taken into account in the  
*computer-generated*  
ICYNENE LD-C-50 PART B

Not Available

|          |     |
|----------|-----|
| Material | CPI |
|----------|-----|

\* CPI - Chemwatch Performance Index  
A: Best Selection  
B: Satisfactory; may degrade after 4 hours continuous immersion  
C: Poor to Dangerous Choice for other than short term immersion  
NOTE

Respiratory protection:

Where the concentration of gas/particulates in the breathing zone, approaches or exceeds the "Exposure Standard" (or ES), respiratory protection is required.  
Degree of protection varies with both face-piece and Class of filter; the nature of protection varies with Type of filter.

| Required Minimum Protection Factor | Half-Face Respirator | Full-Face Respirator | Powered Air Respirator   |
|------------------------------------|----------------------|----------------------|--------------------------|
| up to 10 x ES                      | AK-AUS P2            | -                    | AK-PAPR-AUS / Class 1 P2 |
| up to 50 x ES                      | -                    | AK-AUS / Class 1 P2  | -                        |
| up to 100 x ES                     | -                    | AK-2 P2              | AK-PAPR-2 P2 ^           |

^ - Full-face  
A(All classes) = Organic vapours, B AUS or B1 = Acid gasses, B2 = Acid gas or hydrogen cyanide(HCN), B3 = Acid gas or hydrogen cyanide(HCN), E = Sulfur dioxide(SO2), G = Agricultural chemicals, K = Ammonia(NH3), Hg = Mercury, NO = Oxides of nitrogen, MB = Methyl bromide, AX = Low boiling point organic compounds(below 65 degC)

SECTION 9 Physical and chemical properties

| Information on basic physical and chemical properties |                         |                                                |               |
|-------------------------------------------------------|-------------------------|------------------------------------------------|---------------|
| <b>Appearance</b>                                     |                         |                                                |               |
| White liquid                                          |                         |                                                |               |
| <b>Physical state</b>                                 | Liquid                  | <b>Relative density (Water = 1)</b>            | 1.1           |
| <b>Odour</b>                                          | Ammonia-Like and Slight | <b>Partition coefficient n-octanol / water</b> | Not Available |
| <b>Odour threshold</b>                                | Not Available           | <b>Auto-ignition temperature (°C)</b>          | Not Available |
| <b>pH (as supplied)</b>                               | 11                      | <b>Decomposition temperature</b>               | Not Available |
| <b>Melting point / freezing point (°C)</b>            | Not Available           | <b>Viscosity (cSt)</b>                         | Not Available |
| <b>Initial boiling point and boiling range (°C)</b>   | Not Available           | <b>Molecular weight (g/mol)</b>                | Not Available |
| <b>Flash point (°C)</b>                               | Not Available           | <b>Taste</b>                                   | Not Available |
| <b>Evaporation rate</b>                               | Not Available           | <b>Explosive properties</b>                    | Not Available |
| <b>Flammability</b>                                   | Not Available           | <b>Oxidising properties</b>                    | Not Available |
| <b>Upper Explosive Limit (%)</b>                      | Not Available           | <b>Surface Tension (dyn/cm or mN/m)</b>        | Not Available |
| <b>Lower Explosive Limit (%)</b>                      | Not Available           | <b>Volatile Component (%vol)</b>               | Not Available |
| <b>Vapour pressure (kPa)</b>                          | Not Available           | <b>Gas group</b>                               | Not Available |
| <b>Solubility in water (g/L)</b>                      | Miscible                | <b>pH as a solution(1%)</b>                    | Not Available |
| <b>Vapour density (Air = 1)</b>                       | Not Available           |                                                |               |

SECTION 10 Stability and reactivity

|                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity:</b>                                                                                                                                                                   |
| See section 7                                                                                                                                                                        |
| <b>Chemical stability:</b>                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>• Presence of incompatible materials.</li> <li>• Product is considered stable.</li> <li>• Hazardous polymerisation will not occur.</li> </ul> |
| <b>Possibility of hazardous reactions:</b>                                                                                                                                           |
| See section 7                                                                                                                                                                        |
| <b>Conditions to avoid:</b>                                                                                                                                                          |
| See section 7                                                                                                                                                                        |
| <b>Incompatible materials:</b>                                                                                                                                                       |
| See section 7                                                                                                                                                                        |
| <b>Hazardous decomposition products:</b>                                                                                                                                             |
| See section 5                                                                                                                                                                        |

SECTION 11 Toxicological information

| Information on toxicological effects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhaled:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| The material is not thought to produce either adverse health effects or irritation of the respiratory tract following inhalation (as classified by EC Directives using animal models). Nevertheless, adverse systemic effects have been produced following exposure of animals by at least one other route and good hygiene practice requires that exposure be kept to a minimum and that suitable control measures be used in an occupational setting.<br>Inhalation hazard is increased at higher temperatures.                                                                                                                                                                                              |
| <b>Ingestion:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Accidental ingestion of the material may be damaging to the health of the individual.<br>Nonionic surfactants may produce localised irritation of the oral or gastrointestinal mucosa and induce vomiting and mild diarrhoea.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Skin Contact:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| The material is not thought to produce adverse health effects or skin irritation following contact (as classified by EC Directives using animal models). Nevertheless, good hygiene practice requires that exposure be kept to a minimum and that suitable gloves be used in an occupational setting.<br>One of the mechanisms of skin irritation caused by surfactants is considered to be denaturation of the proteins of skin. It has also been established that there is a connection between the potential of surfactants to denature protein in vitro and their effect on the skin. Nonionic surfactants do not carry any net charge and, therefore, they can only form hydrophobic bonds with proteins. |
| <b>Eye:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Although the liquid is not thought to be an irritant (as classified by EC Directives), direct contact with the eye may produce transient discomfort characterised by tearing or conjunctival redness (as with windburn).

Some nonionic surfactants may produce a localised anaesthetic effect on the cornea; this may effectively eliminate the warning discomfort produced by other substances and lead to corneal injury. Irritant effects range from minimal to severe dependent on the nature of the surfactant, its concentration and the duration of contact. Pain and corneal damage represent the most severe manifestation of irritation.

**Chronic:**

Limited evidence suggests that repeated or long-term occupational exposure may produce cumulative health effects involving organs or biochemical systems. Prolonged or repeated skin contact may cause degreasing with drying, cracking and dermatitis following.

| TOXICITY                        | IRRITATION    |
|---------------------------------|---------------|
| <b>ICYNENE LD-C-50 PART B</b>   |               |
| Dermal (Other) LD50: 2111 mg/kg | Not Available |

|                                                                          |                                      |                                  |                |
|--------------------------------------------------------------------------|--------------------------------------|----------------------------------|----------------|
| <b>ICYNENE LD-C-50 PART B</b>                                            |                                      |                                  |                |
| No significant acute toxicological data identified in literature search. |                                      |                                  |                |
| <b>Acute Toxicity:</b>                                                   | Not Applicable                       | <b>Carcinogenicity:</b>          | Not Applicable |
| <b>Skin Irritation/Corrosion:</b>                                        | Skin Corrosion/Irritation Category 2 | <b>Reproductivity:</b>           | Not Applicable |
| <b>Serious Eye Damage/Irritation:</b>                                    | Eye Irrit. 2                         | <b>STOT - Single Exposure:</b>   | Not Applicable |
| <b>Respiratory or Skin sensitisation:</b>                                | Not Applicable                       | <b>STOT - Repeated Exposure:</b> | Not Applicable |
| <b>Mutagenicity:</b>                                                     | Not Applicable                       | <b>Aspiration Hazard:</b>        | Not Applicable |

**CMR STATUS**

**SECTION 12 Ecological information**

**Toxicity**

For surfactants:

**Environmental fate:**

Octanol/water partition coefficients cannot easily be determined for surfactants because one part of the molecule is hydrophilic and the other part is hydrophobic. Consequently they tend to accumulate at the interface and are not extracted into one or other of the liquid phases. As a result surfactants are expected to transfer slowly, for example, from water into the flesh of fish.

**Persistence and degradability**

|                   |                                |                         |
|-------------------|--------------------------------|-------------------------|
| <b>Ingredient</b> | <b>Persistence: Water/Soil</b> | <b>Persistence: Air</b> |
| Not Available     | Not Available                  | Not Available           |

**Bioaccumulative potential**

|                   |                        |
|-------------------|------------------------|
| <b>Ingredient</b> | <b>Bioaccumulation</b> |
| Not Available     | Not Available          |

**Mobility in soil**

|                   |                 |
|-------------------|-----------------|
| <b>Ingredient</b> | <b>Mobility</b> |
| Not Available     | Not Available   |

**SECTION 13 Disposal considerations**

**Waste treatment methods**

**Product / Packaging disposal:**

Legislation addressing waste disposal requirements may differ by country, state and/ or territory. Each user must refer to laws operating in their area. In some areas, certain wastes must be tracked. A Hierarchy of Controls seems to be common - the user should investigate:

- Reduction

**SECTION 14 Transport information**

|                             |
|-----------------------------|
| <b>Labels Required:</b>     |
| <b>Marine Pollutant: NO</b> |

|                                                                                           |
|-------------------------------------------------------------------------------------------|
| <b>HAZCHEM:</b>                                                                           |
| <b>Land transport (ADG): NOT REGULATED FOR TRANSPORT OF DANGEROUS GOODS</b>               |
| <b>Air transport (ICAO-IATA / DGR): NOT REGULATED FOR TRANSPORT OF DANGEROUS GOODS</b>    |
| <b>Sea transport (IMDG-Code / GGVSee): NOT REGULATED FOR TRANSPORT OF DANGEROUS GOODS</b> |

**SECTION 15 Regulatory information**

**Safety, health and environmental regulations / legislation specific for the substance or mixture**

**SECTION 16 Other information**

**Other information**

Classification of the preparation and its individual components has drawn on official and authoritative sources as well as independent review by the Chemwatch Classification committee using available literature references.

A list of reference resources used to assist the committee may be found at:  
[www.chemwatch.net/references](http://www.chemwatch.net/references)

The (M)SDS is a Hazard Communication tool and should be used to assist in the Risk Assessment. Many factors determine whether the reported Hazards are Risks in the workplace or other settings. Risks may be determined by reference to Exposures Scenarios. Scale of use, frequency of use and current or available engineering controls must be considered.

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# Acoustics

**Product:** The Icynene Insulation System®

## Sound Transmission Class (STC)

**Testing Organization:** Gold Bond Building Products  
Research Center

**Test Method:** ASTM E90-83, E413  
Standard test method for the laboratory measurement of airborne sound transmission loss of building partitions.

**Date:** May 1988

**Result:** STC-37  
Wall assembly consisting of:  
- 2x4 wood studs  
- 3.5" Icynene®  
- 1 layer of 5/8" gypsum wallboard on each side

## Noise Reduction Co-efficient (NRC)

**Testing Organization:** Gold Bond Building Products  
Research Center

**Test Method:** C423-84  
Standard method for the measurement of sound absorption.

**Date:** May 1988

**Result:** NRC – 0.7

**Comments:** The Icynene Insulation System® has an added advantage over other similarly rated materials, in that it air seals the structure when it is applied. The above tests do not allow for loss of rating due to air leakage.



# ICYNENE®

SPRAY FOAM  
INSULATION

## All Insulation Materials are NOT created equal

| Icynene                                                                           |          | Conventional Insulation                                                            |                                                                                     |
|-----------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | Benefits |  |  |
| Icynene                                                                           |          | Fiberglass                                                                         | Cellulose                                                                           |
| Soft sponge                                                                       |          | Feels like                                                                         | Scratchy wool                                                                       |
| ✓                                                                                 |          | Meets Air Barrier Requirements                                                     | Paper litter                                                                        |
| ✓                                                                                 |          |                                                                                    |                                                                                     |
| ✓                                                                                 |          |                                                                                    | ✓                                                                                   |
| ✓                                                                                 |          | ✓                                                                                  |                                                                                     |
| ✓                                                                                 |          |                                                                                    |                                                                                     |
| ✓                                                                                 |          |                                                                                    |                                                                                     |



**ICYNENE®**



## ICYNENE Snapshot



**MORE  
ENERGY  
EFFICIENT**

**ICynene can save you up to 50% off your monthly energy bills.**

**How?** It quickly expands when sprayed to form an air barrier against random air leakage.



**HEALTHIER**

**ICynene is sprayed into walls, ceilings, floors and attics using water-blown technology. ICynene products do not use synthetic blowing agents, such as HFCs, or ozone-depleting substances.**

ICynene helps improve indoor air by sealing out airborne moisture and pollution, including pollens and dust that can trigger allergies or asthma attacks.



**QUIETER  
AND MORE  
COMFORTABLE**

**ICynene can help minimise sound that travels in air. Its sound insulating qualities are perfect for dampening unwanted sounds from plumbing, theatre rooms or street noise.**

ICynene makes your home more comfortable by reducing air leaks, which can cause hot or cold spots and drafty areas.



**MORE  
VERSATILE**

**ICynene lets you effectively insulate and air-seal where some other insulations can't!**

It seals all spaces and sticks to surfaces so it's great for seeping into unusual or hard-to-insulate places like around doors and windows, over garages, in crawlspaces, cathedral ceilings, around electrical boxes and plumbing.



**[www.icynene.com.au](http://www.icynene.com.au) - Australian and New Zealand mini site**

**[www.icynene.com](http://www.icynene.com) - Canadian full site**

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**Era**  
Polymers